

Assessment 2: Analysis Report

AM811001: Analytics for Managers

**Use of Data Analytics for Data Driven Decision Making in
Lending Club**

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1.0 Acronyms Used

LC – Lending Club

P2P – Peer to Peer

DDDM -Data Driven Decision Making

DTI – Debt to income

2.0 Introduction

Data is one of the most potent weapons any industry can use to analyse past, present and future trends in order to compete in the challenging business environment [REDACTED]. Therefore, data analysis has become popularised as a new strategy in the contemporary business context, which converts raw data to meaningful visualisation and provides meaningful interpretations to identify ongoing and future business issues/challenges. Moreover, logical data analytical prediction creates more protection for the organisation, allowing it to be ready for future potential challenges and overcome them with appropriate predefined solutions [REDACTED]. Therefore, this novel strategy is now being used by many organisations rather than traditional decision-making methods to compete in complex business issues and find new avenues for solving business problems efficiently and effectively [REDACTED].

Accordingly, the financial sector is one of the leading industries that use data analytics for decision-making in the contemporary era. Fintech is a novel financial, technological concept that challenges the demanding banking industry by providing simplicity and convenience to customers, influencing the use of other financial applications to cater to banking needs [REDACTED] (2023). Moreover, Fintech facilitates many financial services and provides opportunities across the globe, especially for customers who are in remote areas (Utami & Supriadi, 2023).

Lending Club is an organisation incorporated in 2006 in California, USA, with the intention of helping people with their financial needs; it provides loans to individual borrowers while opening an avenue for investors to invest in the borrowings (Lending Club, 2024). Therefore, this mechanism is called peer-to-peer lending (P2P), and LC acts as a connecting platform for both the borrower and investors (Lending Club, 2024). Moreover, LC evaluates the credibility of the applicants and passes the findings to investors for decision-making on their investment. LC only makes interest and fee income from borrowers and pays back investment returns to the investors on a monthly basis (Lending Club, 2024). Besides, this Fintech Lending and investing application was acquired by Radius Bank and contributed to the national economy of the USA by enhancing its lending portfolio to five billion US dollars by the end of the first half of 2014 (Lending Club, 2024).

Since this Lending Club provide lending from the investor's capital, it is the responsibility of the organisation to pay back interest and capital on a timely basis to protect the investor's interest and keep optimum liquidity in the business. Therefore, it is important to identify trustworthy

prospective borrowers to ensure the timely collection of loan proceeds to safeguard investor interest (Ahmed et al., 2021). Therefore, data analytics is prominent in identifying prospective borrowers and finding new lending opportunities for Lending Club to maximise profits and investor revenue.

Therefore, this report discusses two burning issues of Lending Club: Identifying key factors that influence the credit risk associated with fintech lending for LC and Identifying factors that drive investor interest in a quality lending portfolio within the framework of the data analysis. Moreover, this study used descriptive, diagnostic, predictive, and prescriptive data analysis methods to find solutions for business problems.

Section three of the report provides an insightful description of the data set and its attributes. In contrast, section four critically discusses the framed business problems and illustrates how data analytics provide data-driven solutions for contemporary business issues. Finally, the conclusion summarises the report's findings to complete the report.

3.0 Data Description

The data set is a collection of data that consists of columns and rows, where columns represent the variables of the data set while rows represent the related data to one particular data variable [REDACTED]. This data is mainly categorised as numerical and categorical data, which are further divided into four main categories: nominal, ordinal, interval, and ratio. Identification of the variables and a clear understanding of the data set variables influence the use of correct data analysis methods to accurately reflect the data problems of a contemporary business context [REDACTED].

This database is all about the lending database of Lending Club, which is found on the Kaggle website [REDACTED]), holds issued, settled, current, late and non-performing loans for the period from 2007 to 2018. Over 1.048 Mn rows and 151 data variables were found in the database, and all the data types were found there.

Due to the comprehensiveness of the dataset, data cleaning was challenging, and the data set was cleaned with the anticipation of finding meaningful interpretations for the data problems [REDACTED], 2014). Consequently, data was filtered to get the latest calendar year data for 2017. Therefore, the data had for year 2018 which for a few months, was also removed. Accordingly, rows were reduced to 500K, and further cleaning was performed thereafter by carefully analysing

all the variables and reducing to fifty-eight variables (columns). Some empty rows were filled with zero, and some data rows were removed entirely, which did not have data for important data variables. Focusing accurate analysis results for regression and Rapid minor, data was cleaned aiming the missing values in the crucial numerical data fields. This exercise was performed using pivot tables, Excel functions, and formulas to get the final data set with 212,270 data rows. Considering the requirement of analysis, new columns were added to the dataset, such as joint annual income, joint DTI ratio, average FICO score (credit rating) and monthly loan instalment. After all the cleaning activities, the data set was ready for analysis, and Figure 1 illustrates the data variables and descriptions of the cleaned data set. Moreover, data coding was done using the database for Rapid Minor; refer to Figure 2 below.

Figure 1 - Data variables of a cleaned dataset

	Variable Name	Data Type	Data Measurement type	Description
1	addr_state			
2	annual_inc			
3	annual_inc_joint			
4	application_type			
5	avg_cur_bal			
6	delinq_2yrs			
7	delinq_amnt			
8	dti_joint			
9	earliest_cr_line			
10	emp_length			
11	emp_title			
12	FICO Score			
13	funded_amnt_inv			
14	grade			
15	home_ownership			
16	id			
17	installment			
18	int_rate			
19	Investor Status			
20	issue_d			
21	last_credit_pull_d			
22	last_pymnt_amnt			
23	last_pymnt_d			
24	loan_amnt			

25	loan_status			
26	mort_acc			
27	mths_since_last_delinq			
28	mths_since_recent_bc_dlq			
29	next_pymnt_d			
30	num_accts_ever_120_pd			
31	num_actv_bc_tl			
32	num_bc_sats			
33	num_bc_tl			
34	num_sats			
35	num_tl_120dpd_2m			
36	num_tl_30dpd			
37	num_tl_90g_dpd_24m			
38	open_acc			
39	out_prncp			
40	out_prncp_inv			
41	Present loan installments (Ex. Mortgage Loans)			
42	pub_rec			
43	pub_rec_bankruptcies			
44	purpose			
45	sub_grade			
46	term			
47	tot_cur_bal			
48	tot_hi_cred_lim			
49	total_acc			
50	total_bal_ex_mort			
51	total_bc_limit			
52	total_il_high_credit_limit			
53	total_pymnt			
54	total_pymnt_inv			
55	total_rec_int			
56	total_rec_late_fee			
57	total_rec_prncp			
58	verification_status			

Figure 2 - Data Coding

Sub_grade	Sub_grade coding
A1	1
A2	2
A3	3
A4	4
A5	5
B1	1
B2	2
B3	3
B4	4
B5	5
C1	1
C2	2
C3	3
C4	4
C5	5
D1	1
D2	2
D3	3
D4	4
D5	5
E1	1
E2	2
E3	3
E4	4
E5	5
F1	1
F2	2
F3	3
F4	4
F5	5
G1	1
G2	2
G3	3
G4	4
G5	5

Investor Status	
Investor Status	coding
Full	1
Partially	0

Grade	Grade coding
A	1
B	2
C	3
D	4
E	5
F	6
G	7

Employment Length	Employment Length Coding
< 1 year	0.5
1 year	1
10+ years	10.5
2 years	2
3 years	3
4 years	4
5 years	5
6 years	6
7 years	7
8 years	8
9 years	9

Home_ownership	
Home_ownership	coding
OWN	1
MORTGAGE	2
RENT	3

verification_status	verification_status coding
Verified	1
Source Verified	2
Not Verified	3

Loan Issued data	
Loan Issued data	coding
Apr-17	042017
May-17	052017
Jun-17	062017
Jul-17	072017
Aug-17	082017
Sep-17	092017

loan_status	loan_status coding
Current	1
Fully Paid	2
In Grace Period	3
Late (16-30 days)	4
Late (31-120 days)	5
Default	6
Charged Off	7

4.0 Data-driven decision-making for lending institutions

With the acceleration of digitalisation, new fintech solutions influence banking operations. Fintech lending is more popularised with automated data-driven decision-making (DDDM) using even machine learning algorithms (██████████, 2023). In context, DDDM has been popularised as a new strategy among organisations, which promotes more precisely identifying opportunities and business issues (██████████, 2020). In view of these benefits, lending organisations started aggressively using this new strategy in the competitive business environment, considering its benefits. In contrast, lending is a profitable business to lending organisations, but there is a high tendency to lose the revenue due to bad loans in the lending portfolio. Therefore, data-related findings support identifying lending risk, profitable products, quality customer segments, new product innovations, and market demands to compete with rivals in challenging environments (Gul & Al-Faryan, 2023). Therefore, two critical data problems related to LC are identified and analysed based on data analytics techniques to find the root cause of the problem and provide solutions using DDDM.

4.1 Identify key factors that influence the credit risk associated with fintech lending for LC

Even though Fintech lending is a novel concept in the financial industry, lending is a traditional fund-based financial product for lending organisations. Since this LC platform influences with Peer to Peer (P2P) lending mechanism, supply and demand are created through the Internet, and investors fund loans without collateral even without knowing the borrowers (Serrano-Cinca et al., 2015). Therefore, it has a high tendency to default on loans by the borrowers. Thus, loans not paid for a considerable time frame are considered bad loans, and LC sets off those loans, causing investors a loss of capital investment and interest income. Since fintech solutions are widely spread over new financial solutions, it is important to promptly secure the investor's interest, addressing Fintech-related issues (Reddy & Gopalaraman, 2016). Moreover, the loss of capital causes financial losses to investors and a loss of confidence decrease the interest in Fintech applications. In contrast, this loss of stakeholders' interest may even lead to an impact on the profitability of the LC.

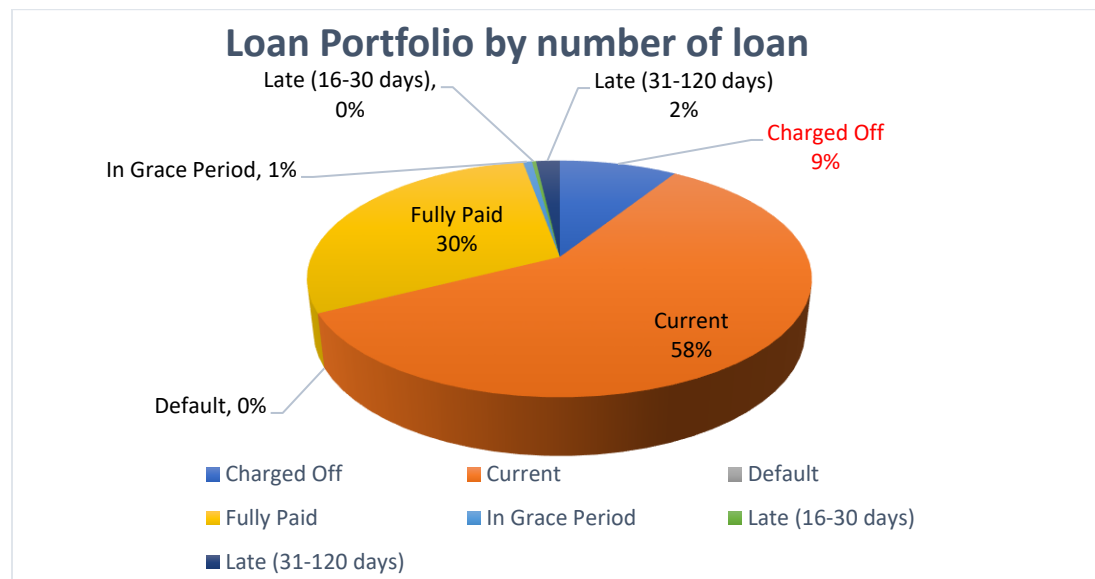
Using data analytics to illustrate this lending problem provides statistical interpretations that provide a measurable understating of the problem statement and allows logical decision-making (Marikar & Bandara, 2020). Moreover, by using data analytics methodologies, it is essential to identify the variable's impact on influencing the credit risk of the borrowers to identify prospective borrowers through the Fintech lending platform. Accordingly, Tabel 1 illustrates the list of variables used to identify the data problem.

Table 1- Data variables used for analysis

	Variable Name	Data Type	Data Measurement type	Description
1	loan_status			
2	loan_amnt			
3	annual_inc			
4	annual_inc_joint			
5	purpose			
6	verification_status			
7	dti_joint			
8	avg_cur_bal			
9	mort_acc			
10	FICO Score			
11	emp_length			
12	funded_amnt_inv			

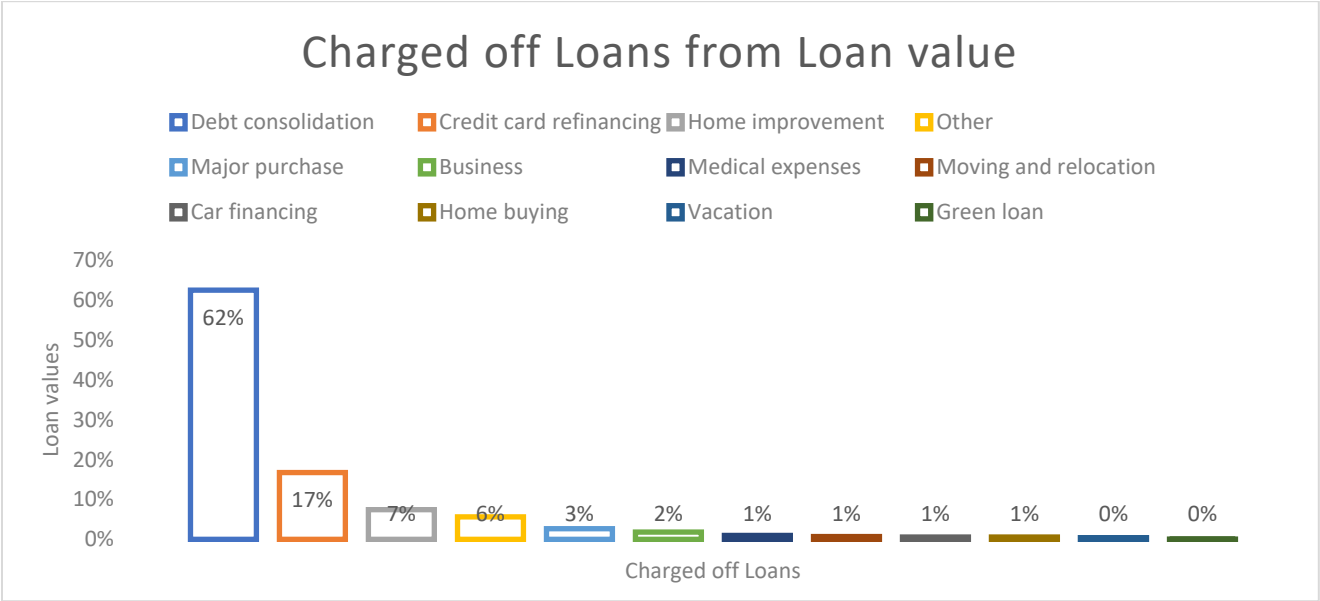
Initially, descriptive analysis was performed over the loan portfolio to identify the distribution of the loans considering the loan status (Figure 3) and identified that there is a variation in the charged-off loans considered as bad loans settled by lending club carried 9% from the loan portfolio which is above the industry average. (Lending Club, 2024). Moreover, loans segmented as default, which is 0%, are the loans of deceased borrowers in the litigation process to settle the dues from the borrower's estate (2024).

Figure 3 - Spread of loans in LC



Only selected the 9% charged off loans for further analysis as this is the segment provides the financial loss to LC. Accordingly, Figure 4 below represents the loan purpose of the charge of loans and identified that 62% of the charged-off loans are obtained for debt consolidation, and the lowest charged-off loans are obtained for vacation loans and green loans. Therefore, the target segment is further narrowed down, to debt consolidation, green and vacation loans studying why the borrowers do not service debt consolidation loans compared to green and vacation loans. Thus, Green and Vacation loans are also further analysed to check what positive factors influence the recovery of loans without dues and how the LC can apply the same strategies to the debt consolidation loans to minimise the credit risk to secure investor interest.

Figure 4 - Charged off loans from loan value



According to Serrano-Cinca et al. (2015), annual income is a substantial variable that showcases borrowers' credibility and ability to make satisfactory repayments. Accordingly, Figure 5 below explains how the borrower's annual income influences the loan repayment ability and reduces the opportunity for loans to be transferred to the non-performing section.

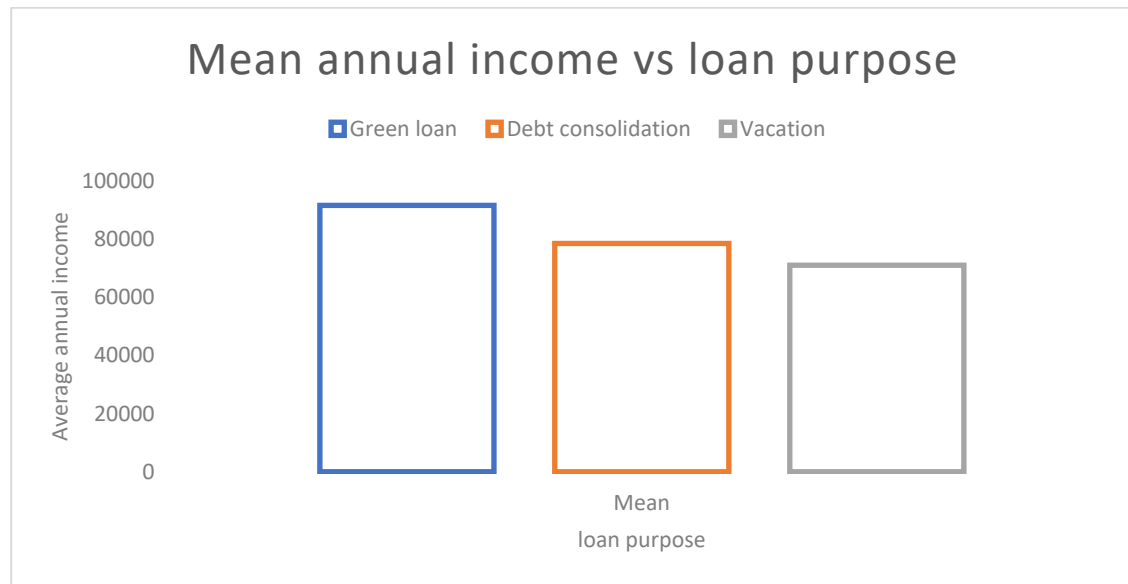
Figure 5 - Loan repayments over the income of the borrowers

[a bar chart was here]

Figure 6 - Descriptive analysis of annual income of the borrower vs loan purpose

Annual income vs loan purpose			
	Debt consolidation	Green loan	Vacation
Mean	78404.47431	91513.88889	70930.33523
Standard Error	415.6423792	14960.94032	3267.682197
Median	70000	75000	60000
Mode	60000	40000	50000
Standard Deviation			
Sample Variance			
Kurtosis			
Skewness			
Range			
Minimum			
Maximum			
Sum			
Count	11257	18	176

Figure 7 - Annual income of the borrower vs loan purpose



Further, to analyse the income of the borrower and three selected loan categories, Debt consolidation, Green loan and Vacation loan, descriptive analysis was performed as per Figure 6 above and claimed the statement of Serrano-Cinca et al. (2015), showcasing green loans which are paid satisfactory due to high income level of the borrowers compared to the Debt consolidation and vacation loans. However, this illustration does not clearly define why vacation loans are servicing better than debt consolidation loans. Therefore, it was encouraged to further analysis the behaviour using Anova single factor diagnostic analysis on these three categorical and numerical variables as Anova confirmed the statistical significance of the variables.

Figure 8 - Diagnostic Anova single factor analysis - Annual income vs loan purpose

Annual income vs loan purpose

Anova: Single Factor

SUMMARY

Groups	Count	Sum	Average	Variance
Debt consolidation	11257			
Green loan	18			
Vacation	176			

ANOVA

Source of Variation	SS	df	MS	F	P-value	F crit
Between Groups	12823504540	2	6411752270	3.293419	0.037162	2.996516
Within Groups	2.22874E+13	11448	1946837737			
Total	2.23002E+13	11450				

Figure 8 confirms that the relationship between the three selected loan purposes and the borrower's annual income is statistically significant as the P value is below the alfa 0.05, which rejects the null hypothesis of "There is no relationship between annual income and loan purpose". In contrast, Reddy and Gopalaraman (2016) emphasise that there is a lower probability of defaulting on loans if the income sources are verified, which influenced to perform the diagnostic chi-square analysis over two categorical variables of income verification and income slabs to analyse the relationship between variables. (Figure 9 illustrates the annual income verification status of the LC)

Figure 9 - Income verification status

The income verification status.
• Verified – Income source documents were obtained for verification. • Source verified - Call the borrower's company to confirm that the employee works there. • Not verified – Some income levels are not targeted to verify, and none of the above verifications were obtained.

Figure 10 - Chi-square analysis - annual income vs annual income verification

Observed Values				
Annual Income	Not Verified	Source Verified	Verified	Grand Total
1000-10999				3879
11000-20999				4376
21000-30999				2056
31000-40999				1140
Grand Total	2732	5036	3683	11451

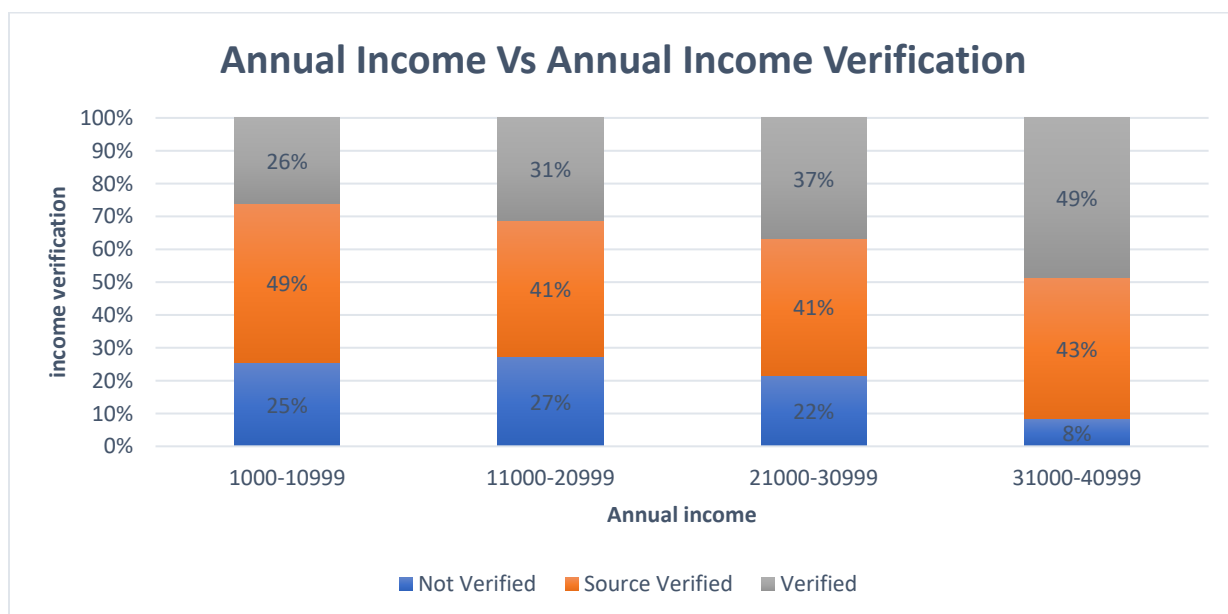
Expected Values				
Annual Income	Not Verified	Source Verified	Verified	Grand Total
1000-10999				3879
11000-20999				4376
21000-30999				2056
31000-40999				1140
Grand Total	2732	5036	3683	11451

Chi-square p-value

8.7762E-69

In reference to the Chi-square analysis as per Figure 10 above, P value is below the alpha value of 0.05, emphasise the relationship between the annual income and the income verification is statistically significant, elaborate that for applicants where the annual income is high, there is a high tendency to verify the annual income and vice versa (Figure 11). Therefore, these findings relate to the previous analysis in Figure 7, illustrating a high tendency of payback loans by the borrowers with good income levels which is also verified by LC ([REDACTED] 2016).

Figure 11 - Annual Income Vs Annual Income Verification



Moreover, another diagnostic chi-square analysis was performed with two categorical variables, loan purpose and annual income, which is also confirmed that the two variables are statistically significant, referring to Figure 12 as the P value is below the alpha value of 0.05 and Figure 13 also provides the graphical presentation over the relationship. Therefore, it confirmed that the green loan borrowers' annual income is more verified than the Debt consolidation and Vacation loans, which influenced the payback loans. Besides, where the income is not verified or only sourced verified in most of the debt consolidation loans, which illustrated that borrowers of the debt consolidation loans had provided false income details and which influenced the defaulting of the loans of debt consolidation (Mian & Sufi, 2015). However, the factors affecting vacation loan defaulting risk are still questionable.

Figure 12 - Chi-square analysis over Loan purpose vs income verification

Observed values				
Loan Purpose	Not Verified	Source Verified	Verified	Grand Total
Debt consolidation	■	■	■	11257
Green loan	■	■	■	18
Vacation	■	■	■	176
Grand Total	2732	5036	3683	11451

Expected values				
Row Labels	Not Verified	Source Verified	Verified	Grand Total
Debt consolidation	■	■	■	11257
Green loan	■	■	■	18
Vacation	■	■	■	176
Grand Total	2732	5036	3683	11451
Chi-square p-value	0.030860494			

Figure 13 - Purpose of Loan vs Annual income verification (proportion-wise)

[A bar chart here]

The debt-to-income (DTI) ratio is one of the crucial lending evaluators for unsecured digital loans, where lower DTI is preferred for quality lending, showcasing a positive correlation between loan defaults has been proven based on a research study on the Indonesian P2P lending market. ■■■■■ (2019). Therefore, it influenced further descriptive analysis of how DTI impacts the United States P2P lending market. (Figure 14)

Accordingly, through the descriptive analysis refer to Figure 14 identified that the average DTI on Debt consolidation loans is the highest compared to the green and vacation loans. Therefore, another diagnostic ANOVA test was performed on the categorical variable (three sub-categories) and numerical variables to find the relationship between the variables. (Figure 16)

Figure 14 - Descriptive analysis of DTI ratio over three loan purposes

	Debt consolidation	Green loan	Vacation
Mean	20.12542951	15.50111111	19.56107955
Standard Error			
Median			
Mode			
Standard Deviation			
Sample Variance			
Kurtosis			
Skewness			
Range			
Minimum			
Maximum			
Sum			
Count	11257	18	176

Figure 15 - Graphical explanation of DTI ratio over three loan purposes

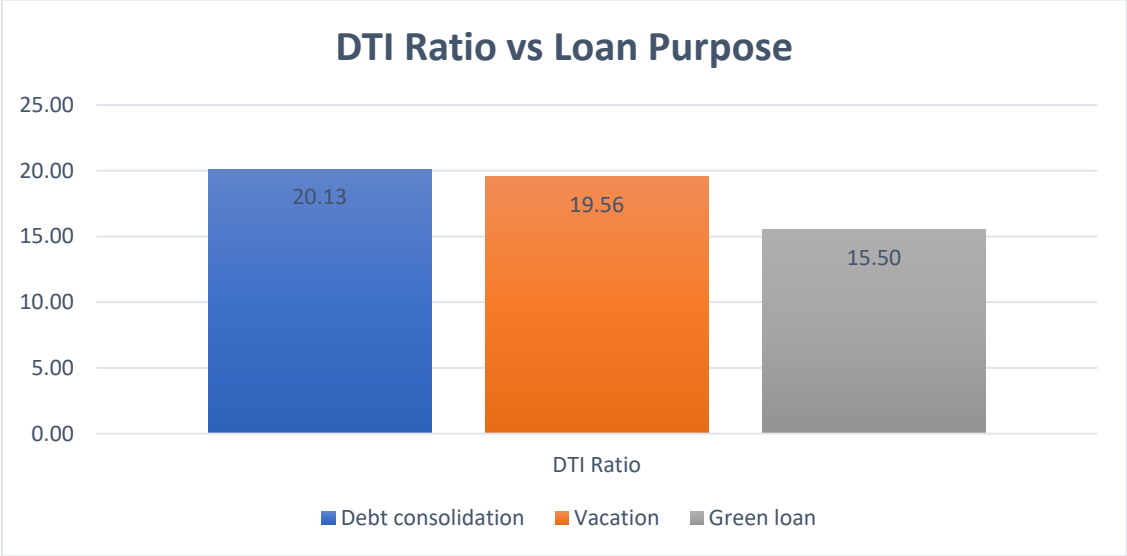


Figure 16 - Diagnostic Anova single factor analysis - DIT ratio over loan purpose

Anova: Single Factor

SUMMARY				
Groups	Count	Sum	Average	Variance
Debt consolidation	11257			
Green loan	18			
Vacation	176			

ANOVA						
Source of Variation	SS	df	MS	F	P-value	F crit
Between Groups	438.0615597	2	219.0307798	3.495727437	0.03035906	2.996516
Within Groups	717294.0146	11448	62.65670987			
Total	717732.0762	11450				

Figure 16 Anova analysis represents that the variables are statistically significant as the p-value is below the alfa value of 0.05. Referring to Figure 15, it is confirmed that there is a high tendency to default on debt consolidation loans compared with Green and vacation loans. However, no significant difference is recognised between debt consolidation and vacation loans. Besides, vacation loans are associated with emotional spending, and the borrower's psychological behaviour also influences payback loans (). Therefore, repayments for loans obtained for different purposes such as vacation, education and medical are also correlated with the psychological factor of the borrower, which this dataset may not answer as some of the human behavioural variables are limited in the dataset (2012).

Finally, correlation analysis (Figure 17) was done on some numerical variables, and a negative, weak correlation between the average account balance and the DTI ratio was found. This emphasises that borrowers who tend to default on loans may have lower credit balances in their savings accounts. Other variables analysed with DTI are not correlated for further interpretations.

Figure 17 - Diagnostic Correlation analysis

	The average current balance of all accounts	No. of mortgage loans	[REDACTED]	[REDACTED]	[REDACTED]	Amount funded by investors	DTI Ratio
The average current balance of all accounts	1						
No. of mortgage loans	0.488078306	1					
[REDACTED]	0.09917239	0.130071	1				
[REDACTED]	0.070870455	0.1647476	0.01390109	1			
[REDACTED]	0.237754186	0.2424563	0.09785487	0.04903015	1		
Amount funded by investors	0.237785785	0.2424955	0.09792162	0.04907188	0.9999952	1	
DTI Ratio	-0.104748605	-0.008183	-0.0528239	0.03723544	0.042102	0.0420666	1

Considering the credit risk involved with fintech P2P lending, LC should focus more on the repayment behaviours of the borrowers and consider more insightful evaluations to select the prospective borrowers ([REDACTED]). Considering only the borrower's income does not reflect the ability to repay, whereas considering the expenses along with the income provides a more meaningful evaluation for lending (Ahmed et al., 2021). In contrast, considering the DTI ratio for lending is more insightful and can build a healthy advances portfolio for LC [REDACTED]

Moreover, Reddy and Gopalaraman (2016) argued that income verification is essential for lending evaluations as there is a high tendency to forge income statements to obtain loans. Therefore, proper income verification provides a quality lending basket as it reduces the credit risk of falling loans to the past due section ([REDACTED]). Further, promoting more Green and Vacation loans, may reduce the non-performing loan ratio of the LC. Customer account balances can also

be checked to find the customer can service the loan without default. As such, if the customer's monthly average account balance is above the proposed loan instalment, it indicates that the customer can service the loan without any financial obligation (██████████).

4.2 Identifying factors that drive investor Interest in a quality lending portfolio

LC influence P2P lending through Fintech applications where investors are the key stakeholders funding the organisation to make a profitable income. Therefore, investor interest is significant, and LC is primarily responsible for safeguarding investors by providing profitable investor opportunities (Lending Club, 2024).

P2P lending is always influenced by the investor's decision-making. Setiawan et al. (2020) argued that investor decision-making is not always logical as decision-making is more considered socialistic factors like the borrower's first impression. Therefore, it is essential to investigate the factors that influence decision-making on P2P lending to protect investor interest and influence investors to make effective decision making on lending investments. Luo et al. (2011) argued that addressing investor problems is more curtail than losing investor interest, which can also negatively impact the business operation of LC.

Moreover, data analytics will play a significant role in solving this problem statement as descriptive analysis can elaborate the historical data and diagnostic analysis can find the root cause of the occurrences while predictive analysis can predict the future trend of the occurrence of the negative implications and further illustrate future prospective investor opportunities. Therefore, the variables used for analysing the caption problem statement are listed below in Table 2

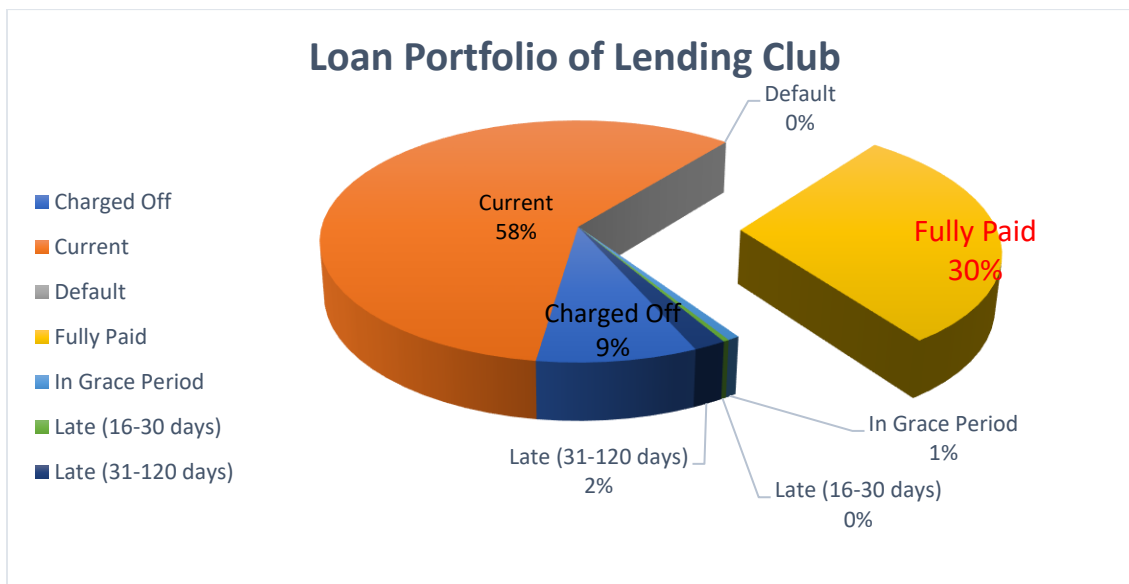
Table 2 - Data variables used for analysis

Variable Name		Data Type	Data Measurement type	Description
1	loan_status			
2	annual_inc			
3	loan_amnt			
4	funded_amnt_inv			
5	annual_inc			
6	FICO Score			
7	emp_length			

8	dti_joint			
9	term			
10	int_rate			
11	Present loan installments (Ex. Mortgage Loans)			
12	avg_cur_bal			
13	tot_cur_bal			
14	num_actv_bc_tl			
15	num_bc_sats			
16	num_bc_tl			
17	tot_hi_cred_lim			
18	total_bal_ex_mort			
19	total_bc_limit			
20	total_il_high_credit_limit			
21	installment			

With the intention of finding what factors influence investors to make lending investments, initially, descriptive analysis was performed to identify the current trends of the business. Considering the term “quality lending portfolio”, only the fully paid loans selected from the loan basket which is identified as 30% of the total loan portfolio (Figure 18).

Figure 18 - Loan portfolio of LC



Therefore, further analysis was performed on fully paid loans to determine the investor's interest in the quality lending portfolio. Accordingly, the income level of the borrowers segmented and Figure 19 bar chat showcase how settled loans are distributed over the borrowers' income. It is found that borrowers who obtain loans and settled are more centric towards low-income earners, and 98% of the loans are obtained by borrowers who earn annual income below \$ 260,000. Moreover, Figure 20 illustrates that 97% of the loan amount is also centric with the same set of borrowers who earn an annual income below \$ 260,0000. Therefore, with the intention of filtering the dataset, focusing on accurate findings and avoiding deviations from mean values, further analysis was carried out based on the borrowers earning annual income below \$ 260,000 [REDACTED] [REDACTED]).

Figure 19 - Number of loans over the annual income of the borrower

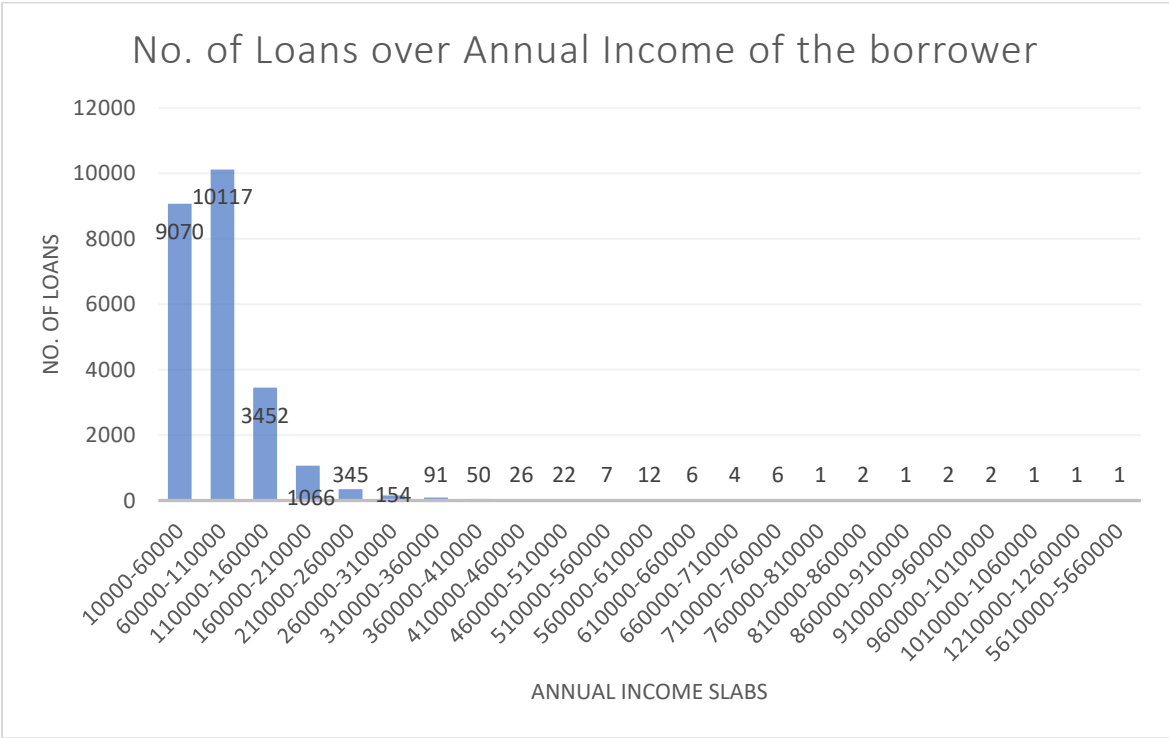
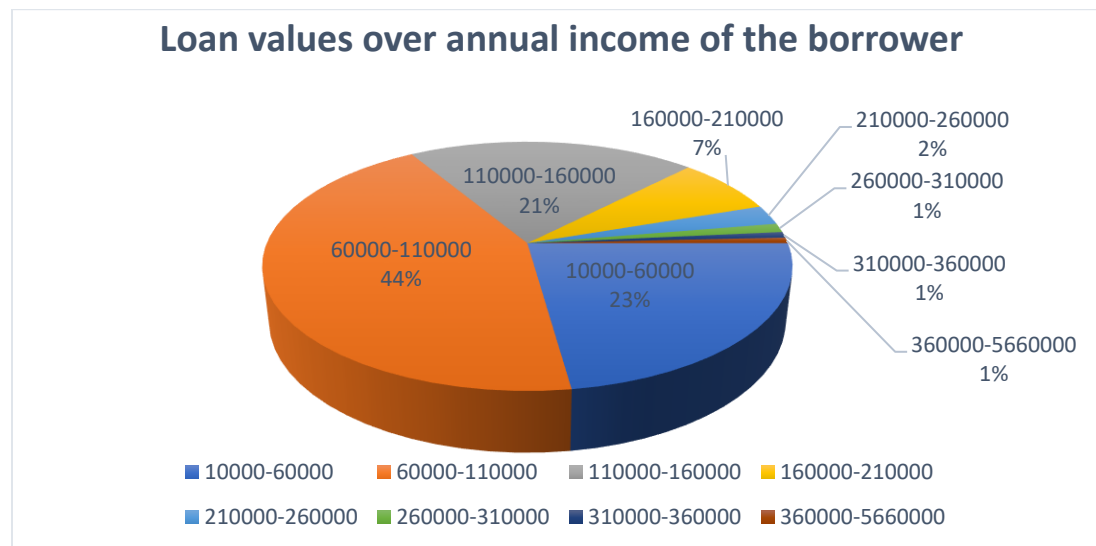


Figure 20 - Loan value over the annual income of the borrower



Diagnostic correlation analysis performed using 17 numerical variables to find the correlation with the “amount funded by investors” (Figure 21). The correlation analysis (Figure 22) identified that the annual income of the borrowers positively correlated with the amount funded by investors, which provides a clear understanding that annual income is one of the key variables of investors’ interest in making investment decision-making (). According to the business model, LC makes monthly capital and interest returns once the borrower makes the monthly instalment (Lending Club, 2024). In context, correlation analysis showcases this relationship as a strong positive correlation with the monthly instalment and the amount funded by the investors. Moreover,) claimed that investors are more keen on profitable investments and higher monthly returns. Even though the DTI ratio and borrowers’ credit rating (FICO score) also positively correlated with the amount funded by the investors, the correlation is negligible and reflects that the investors are not paying more attention to the more critical factors on lending investments due to poor knowledge in the financial literacy. ().

Figure 21 - Correlation analysis over 17 numerical variables

	Amount funded by investors	Annual income of the borrower/s	Average FICO Score	Employment experience	DTI	Loan Term (Months)	Interest Rate on the loan	Monthly loan instalment	Average current balance of all accounts	Total balance of all accounts	No. of active bank cards	Age	Gender	Marital Status	Number of children	Number of pets	Number of siblings
Amount funded by investors	1																
Annual income of the borrower/s	0.530690441	1															
Average FICO Score	0.009933852	0.121738591	1														
Employment experience	0.085778647	0.106962749	0.065078942	1													
DTI	0.078022112	0.094132218	0.138983917	0.064886222	1												
Loan Term (Months)	0.39835281	0.169320244	0.023128009	0.057596154	0.061561765	1											
Interest Rate on the loan	0.154328066	0.111816123	0.498483741	0.036738254	0.192590208	0.339658097	1										
Monthly loan instalment	0.954843031	0.486718198	0.049400432	0.073103457	0.096147346	0.177563653	0.210890325	1									
Average current balance of all accounts	0.245712868	0.426250414	0.127126978	0.1034549	0.050655882	0.12474004	0.093158711	0.203881794	1								
Total balance of all accounts	0.340040609	0.564475546	0.099658552	0.133908504	0.096773885	0.164674797	0.077819869	0.293084764	0.822454316	1							
No. of active bank cards	0.212875658	0.200679188	0.278943178	0.127875493	0.216353242	0.065183711	0.12725309	0.230225426	0.105692305	0.13618971	1						
Age	0.218204932	0.2223724	0.031874739	0.125106193	0.140376164	0.065173686	0.002387296	0.218539722	0.122960814	0.152552209	0.801111767	1					
Gender	0.234354974	0.265712585	0.094011588	0.175838707	0.122774706	0.066482714	-0.06756292	0.224853457	0.033043051	0.208043048	0.619418342	0.810447557	1				
Marital Status	0.367403639	0.604450921	0.186243913	0.160227309	0.109086383	0.162256806	0.128831445	0.315244503	0.77908126	0.98122991	0.174819684	0.228999403	0.290421175	1			
Number of children	0.305018309	0.457552783	0.048064307	0.026269606	0.374650263	0.138737448	0.030846387	0.283770369	0.297094124	0.530835204	0.184210395	0.175990738	0.200200839	0.522368765	1		
Number of pets	0.356088471	0.456422249	0.283970729	0.132902582	0.05510644	0.083906976	0.188729799	0.322980019	0.128497383	0.345608065	0.476088044	0.647066693	0.640560573	0.466191997	0.296599282	1	
Number of siblings	0.233051384	0.399878728	0.075133355	0.001602274	0.409242507	0.107494699	0.039165189	0.207398642	0.231246015	0.433073619	0.044565763	0.073022204	0.114316185	0.43803943	0.885961199	0.174975125	1

Figure 22 - Correlation analysis of 16 numerical variables over “Amount funded by investors.”

[A table here]

The predictive analysis focuses on independent and dependent numerical data variables to identify unexpected business opportunities and threats in business environments (). Therefore, Multi Lenier Regression analysis was performed considering “Amount funded by investors as the dependent (Y) variable and considered all other 16 numerical variables as independent (X) variables (Figure 23).

Figure 23 - Predictive analysis - Multi-Lenier Regression analysis

[A table here]

Multiple R represents the correlation of the variables, which emphasises that the dependent variable is highly correlated with the independent variables. Moreover, R square illustrates that 98% of the variation in investor interest for funding for loans is explained by the 16 variables listed in the regression list in Figure 23. Significance f is zero, below the alfa value of 0.05, illustrating that the variables are statistically significant with the investor funding amount. Apart from the three variables of DTI, the total credit balance, excluding mortgage and total instalment of a high credit/credit limit, are the only independent variables that are not statistically significant with the dependent variable. Therefore, this interpretation confirms that investors' decisions to invest in P2P lending are influenced by many borrower-related details. However, some of these curtail factors have not yet been identified as crucial by investors.

Figure 24 - RapidMiner Predictive clustering analysis

k-Means - Summary

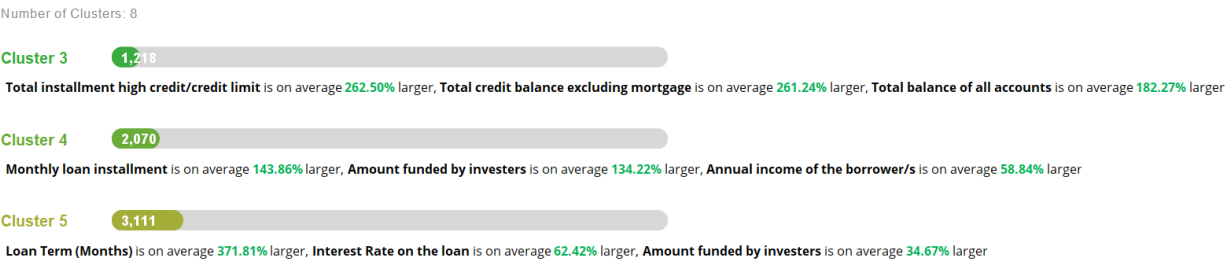


Figure 25 - RapidMiner Centroid Table

k-Means - Centroid Table

Cluster	Amount ...	Annual I...	Averag...	Averag...	DTI	Employ...	Interest...	Loan Te...	Monthly...	No. of a...	Number...	Number...	Total ba...	Total ba...	Total cr...	Total hi...
Cluster 0	10770.675	64354.834	8147.760	697.931	22.480	6.386	13.535	36.112	366.108	5.004	8.575	6.337	99217.565	26001.814	48981.361	130427.527
Cluster 1	17711.323	123239.403	49609.813	739.108	13.318	6.506	10.284	42.386	507.088	2.819	6.685	4.031	424609.677	35952.493	55214.952	485105.796
Cluster 2	7422.037	49336.075	6472.758	703.909	14.713	4.032	13.220	36.129	250.739	2.231	3.803	2.814	44901.035	10488.126	25320.422	59176.763
Cluster 3	21136.628	138380.435	29703.131	722.107	23.135	5.862	12.376	45.253	604.391	3.966	8.619	5.600	416653.546	46072.592	185314.768	483417.394
Cluster 4	31274.191	120031.188	15799.107	720.163	18.426	5.958	13.875	41.435	985.764	4.075	8.342	5.583	179994.197	41675.595	67206.658	234143.854
Cluster 5	18406.905	74424.727	12361.556	714.346	18.697	6.026	16.967	59.931	460.699	3.324	6.445	4.451	120105.688	23050.619	46215.168	152770.424
Cluster 6	9110.637	73463.391	13182.836	765.165	15.725	6.566	7.831	36.471	281.890	2.373	6.840	4.161	120959.831	29443.104	39163.102	170275.416
Cluster 7	17913.029	105114.220	11155.148	725.445	20.134	7.315	11.523	42.149	534.695	7.975	17.366	12.284	208731.833	80156.787	70643.572	302805.907

RapidMionr clustering is more suitable for predictive analysis, which uses machine learning algorithms to categorise data according to the similarities and provide meaningful interpretation, making linkages between variables (RapidMiner Documentation, 2024). Accordingly, RapidMinor clustering analysis also interprets the investor behaviour on investor decisions. As

believed, cluster 4 interprets that Moreover, cluster 5 explains investors' interest in investing in long-term loans, which have high rates. Although the rest of the clusters are not interpreted as only clusters 4 and 5 are related to the data problem

In reference to Setiawan et al. (2020), investor decision-making is mainly based on the return on investment and the risk involved in the investment opportunity. Moreover, investment decision-making is further influenced mainly by three factors: information provided by the lending platform, the first impression of the borrower and social and emotional influences (Setiawan et al., 2020). Besides, more investors are not literate about financial evaluations, and their investment decisions are mainly based on basic evaluation methodologies such as income, savings balance,

employment experience and satisfactory bank card details. However, significantly less consideration is given to the borrower's DTI ratio and FICO credit score (Lending Club, 2024a). Therefore, LC is primarily responsible for filtering prospective borrowers and providing profitable investment opportunities to investors. Thus, it is recommended that LC share the primary knowledge with investors to identify prospective borrowers and make profitable investments.

5.0 Conclusion

Data analytics is a novel strategy that organisations use to be competitive in demanding markets, providing clear and accurate findings to identify and interpret real problems in a business environment. Moreover, it opens new avenues for finding solutions to business issues. Consequently, two data problems of LC are discussed in this report, namely identifying the factors that influence the credit risk in LC and identifying factors that drive investor interest. Therefore, many of the analysis techniques, such as descriptive, diagnostic, and predictive, were used to understand the data problems of LC, and it was found that the findings are valid and need to be actionable to improve revenue and safeguard investor interest.